Adnan Harun Dogan

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Education

Master of Science in Computer Engineering — *Sep 2022 – Jan 2025*
Middle East Technical University (METU) — *Ankara, Turkey*
- **Thesis:** Incorporating Piecewise Linear Functions with Constant Regions in Backpropagation
- **Advisor:** Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas

Bachelor of Science in Computer Engineering — *Sep 2018 – Jul 2022*
Middle East Technical University (METU) — *Ankara, Turkey*

Experience

Graduate Intern (Supervisor: Prof. Dr. Christian Holz) — *Mar 2025 – Oct 2025*
Sensing, Interaction, & Perception Laboratory (SIPLab), ETH Zürich — *Zürich, Switzerland*
- Developed learned frequency-weighted Kalman filters that denoise the innovation spectrum under spectral supervision
- Built EEG multitaper-spectrum deep models (ConvLSTM) for real-time VR cybersickness detection, +30% over baselines

Research Assistant (Supervisor: Prof. Dr. Sinan Kalkan, Assoc. Prof. Dr. Emre Akbas) — *Feb 2023 – Present*
Image Processing and Pattern Recognition Laboratory (ImageLab), METU — *Ankara, Turkey*
- Contributed to Bucketed Ranking Loss, speeding up ranking-based training of DETR/transformer detectors by up to 6x
- Developed optimal-transport (Sinkhorn-Knopp) and ranking-based loss formulations for end-to-end object detection

Undergraduate Intern (Supervisor: Assoc. Prof. Dr. Erol Sahin, Assoc. Prof. Dr. Bugra Koku) — *Oct 2021 – Feb 2022*
Center for Artificial Intelligence and Robotics (ROMER), METU — *Ankara, Turkey*
- Built a self-driving stack for a physical RC car, and trained RL agents for autonomous navigation task in ROS/Gym/PyTorch
- Simulated a Hector-quadrotor swarm in ROS/Gazebo, approximating numerical solvers for multi-agent localization

Undergraduate Intern (Supervisor: Prof. Dr. Mohammad Abdullah Al-Faruque) — *Jun 2021 – Sep 2021*
Cyber-Physical Systems Laboratory, University of California, Irvine — *Irvine, California, USA*
- Built multi-modal deep models for beat-by-beat cardiac and sleep abnormality detection from physiological signals
- Fused time-series and spectral features for arrhythmia classification, competitive in international challenges

Publications

Dogan, A. H., Demirel, B. U., Holz, C. (2026). *Frequency-Weighted Neural Kalman Filters.* **ICRA 2026**. PDF · Code

Demirel, B. U., **Dogan, A. H.**, Rossie, J., Möbus M., Holz, C. (2025). *Beyond subjectivity: Continuous cybersickness detection using eeg-based multitaper spectrum estimation.* **IEEE/TVCG 2025**. DOI · Code

Yavuz, F., Cam B.C., **Dogan, A. H.**, Oksuz, K., Kalkan, S., Akbas, E. (2024). *Bucketed Ranking Loss for Efficient Ranking-based Training of Object Detectors.* **ECCV 2024**. DOI · PDF · Code

Amin M. A., **Dogan, A. H.**, Kuru E. S., Sever Y., Angin P. (Apr 2024). *Misuse Detection and Response for Orchestrated Microservices Based Software.* AINA 2024. PDF · Data

Karagoz, P., Cekinel, R. F., **Dogan, A. H.**, Oktay, B., Ozturk, A. U., Tonay, S. T., Tunel, B. M. (2024). *Enhancing Underground Built Heritage Analysis with Text Mining: A Case Study on Cappadocia.* Book Chapter 2024. PDF

Sever Y., **Dogan, A. H.** (Aug 2023). *A Kubernetes dataset for misuse detection.* ITU Journal 2023. DOI · PDF · Code

Sever, Y., Ekinci, G., **Dogan, A. H.**, Alparslan, B., Gurbuz, A. S., Jabrayilov, V., Angin, P. (Sep 2022). *An Empirical Analysis of IDS Approaches in Container Security.* IEEE/SRMC 2022. Best Paper Award DOI

Demirel, B. U., **Dogan, A. H.**, Al-Faruque, M. A. (2021). *Two Might Do: A Beat-by-Beat Classification of Cardiac Abnormalities using Deep Learning and Domain-Specific Features.* CinC 2021. DOI · PDF · Code

Buyukbas, E. B., **Dogan, A. H.**, Ozturk, A. U., Karagoz, P. (Sep 2021). *Explainability in Irony Detection.* DaWaK 2021. PDF

Dogan, A. H., Dogan, A. (Jun 2021). *An Assembled Deep Learning Approach for Flow Field Prediction.* PDF

Scores, Honors, & Awards

TOEFL iBT: 109/120 (R29 L29 S25 W26), ETS — *Oct 2022*

University Entrance Exam: ranked 1466th of 2.27M candidates, OSYM, Turkey — *Aug 2017*

Technical Skills & Research Interests

Languages & Tools: Python/PyTorch, C/C++, ROS2, Rust
DevOps & HPC: GitHub, Docker, Slurm (Workload Manager), Apptainer
Differentiable Optimization: Combinatorial Optimization, Optimal Transport, Deep State Space Models
Deep Learning: Computer Vision, Self-Supervised Learning, Probabilistic Deep Learning, Deep Reinforcement Learning